NodSense

Made by PassoCrafto

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I. Introduction

Communication is 70% context and only 30% words. For students and professionals moving between cultures, miscommunication can lead to missed opportunities. Our project, NodSense, acts as a 'Cultural Companion'. By analyzing social cues and cultural frameworks, our AI extension provides users with instant insights into why a sentence is phrased a certain way, ensuring that no meaning is lost in translation. This information can be found in [1]

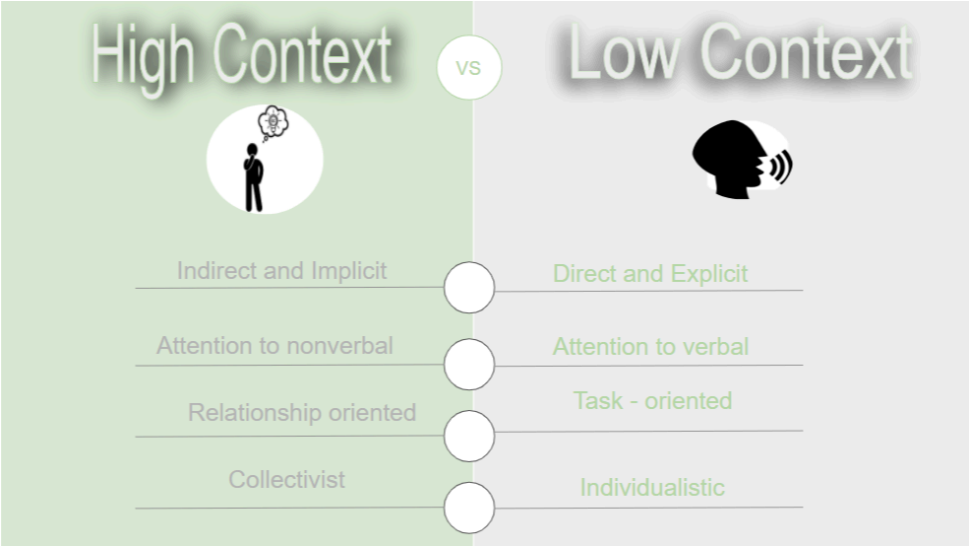

Figure 1: Differences between High and Low context

There are a variety of AI embedded extensions which are now free and available on Chrome, such as Grammarly or Quillbot, and their effectiveness and practical support are undeniable. Regarding the technical architecture, we made a strategic decision to prioritize API-based integration (specifically the Gemini 1.5 Flash API) over the development of a custom fine-tuned model. Developing a specialized model from scratch is a resource-intensive process requiring massive datasets and high-end computational power (GPU hours) that can hinder rapid prototyping. By choosing an API-centric approach, we achieved:

Cost-Efficiency: Maintaining a lean budget by paying only for inference.

High-Performance Reasoning: Accessing state-of-the-art cultural decoding capabilities without the overhead of model maintenance.

Rapid Time-to-Market: Allowing NodSense to reach early adopters quickly and gather the real-world feedback necessary for future iterations.

II. Review of Existing Solutions

Current language and communication tools (e.g., Grammarly, Google Translate, or AI transcription services) primarily focus on grammar correction, literal translation, or basic speech-to-text accuracy. They fail to decode underlying cultural nuances, such as mistaking an Indonesian team member's silence for agreement rather than hesitation, or literally translating a Vietnamese member's "I do not think this one is fine but it is up to you" without capturing its true intent of polite disagreement.

Furthermore, standalone applications or external websites disrupt the user's workflow. Therefore, a seamless, context-aware solution integrated directly into daily collaborative platforms—like our proposed Chrome Extension—is necessary to provide real-time cultural interpretation without adding friction.

III. Methodology

1. System Architecture

– The application is built on an event-driven architecture utilizing the Chrome Extension Manifest V3 framework.

– The content script acts as the client-side module, directly interacting with the Google Meet Document Object Model (DOM) to manage the dynamic user interface and extract raw text.

– A background service worker functions as a local middleware, handling asynchronous tasks and managing secure cross-origin communication with the external language model via RESTful API calls.

Figure 2: Development platform

2. Real-time Data Acquisition

– To capture live captions without causing browser performance degradation, the system implements the MutationObserver interface instead of resource-heavy polling algorithms.

– The algorithm actively monitors specific DOM nodes containing the subtitle data and applies string filtering techniques to continuously extract the speaker's name and the corresponding transcript.

– A debouncing mechanism is applied using a 4000-millisecond timeout to aggregate fragmented text into complete, semantically meaningful sentences before triggering the analysis workflow, significantly reducing unnecessary API requests.

3. Prompt Engineering and LLM Integration

– The core analytical engine relies on the Google Gemini model to perform deep semantic analysis of the captured corporate conversations.

– Strict prompt engineering constraints are applied to force the AI to return data in a structured Raw JSON format rather than unpredictable natural language.

– The extracted data fields include literal translations, cultural nuances, quantified agreement or hesitation percentages, and actionable English reply suggestions.

4. System Optimization and Security Constraints

– To comply with the free-tier API rate limits and prevent HTTP 429 Too Many Requests errors, a state-based throttling mechanism was developed.

– For credential security, the API key is strictly isolated within a local environment configuration file and explicitly excluded from version control, preventing unauthorized access when the source code is published on public repositories.

– This mechanism enforces a mandatory 15-second cooldown period between consecutive API calls, ensuring sustained application stability during extended meeting sessions.

5. Code Description

The codebase is divided into distinct scripts to separate the user interface, live data extraction, and backend API communication logic.

– `manifest.json` (Extension Configuration): This file acts as the primary blueprint for the extension. It declares the Manifest V3 architecture and defines the necessary security permissions, specifically granting host permissions to interact with Google Meet URLs and the Google Generative Language API. It also registers the background service worker and the content scripts.

– `content.js` (Client-Side Logic & UI Management): Injected directly into the Google Meet DOM, this script functions as the frontend engine. It handles three critical tasks:

1. **UI Injection**: Dynamically generates a resizable, draggable, and non-intrusive dashboard overlay on the meeting interface.
2. **Live DOM Observation**: Employs the MutationObserver API to continuously monitor the subtitle containers (.adE6rb) and extract raw text and speaker names in real-time without polling overhead.

3. **Flow Control**: Implements a 4-second debouncing timer to aggregate fragmented speech into complete sentences, alongside a 15-second cooldown variable (isCoolingDown) to prevent API spamming. It uses chrome.runtime.sendMessage to securely pass the extracted text to the background script.

– `background.js` (Service Worker & AI Integration): Operating asynchronously as a local backend, this script is responsible for external communications. It listens for message payloads from content.js, constructs a strict prompt structure, and executes HTTP POST requests to the Gemini 2.5 Flash API. Furthermore, it sanitizes the AI's response (removing Markdown formatting), parses the Raw JSON, calculates the exact hesitation/agreement percentages mathematically, and returns the structured object to the frontend for rendering.

– `styles.css` (Presentation): Styles.css handles the visual hierarchy and CSS properties of the injected dashboard, ensuring high contrast and readability within the Google Meet environment.

IV. Results

1. System Outputs

Upon successful deployment in a live Google Meet environment, the ASEAN Nuance Decoder extension processes real-time captions and consistently outputs a structured, easy-to-read dashboard. For each captured speech segment, the system successfully generates:

– **Deep Semantic Translation (Meaning)**: Bypasses literal translation to decode corporate jargon and reveal the speaker's true intent.

– **Cultural Nuance Detection (Nuance)**: Accurately identifies the underlying tone of the conversation (e.g., polite refusal, indirect disagreement, or genuine enthusiasm).

– **Quantitative Sentiment Metrics**: Visually displays the speaker's stance through calculated agree_percent and hesitate_percent progress bars.

– **Actionable Reply Suggestions**: Generates context-aware, grammatically correct English responses (accompanied by native language translations) to help the user reply instantly.

– **Keyword Highlighting**: Extracts and highlights critical vocabulary from the original transcript, drawing the user's attention to the most important parts of the speech.

2. Practical Benefit & Impact

The implementation of this system yields significant advantages for cross-cultural corporate communications:

– **Bridging Cultural and Language Gaps**: It empowers non-native English speakers and professionals working in diverse environments to "read between the lines," preventing miscommunications caused by indirect phrasing or cultural differences.

– **Real-time Cognitive Support**: By providing instant translation and suggested replies, the tool significantly reduces the user's cognitive load and anxiety, allowing for smoother and more confident interactions in high-pressure meetings.

– **Non-intrusive User Experience**: The dashboard is designed to be fully resizable, draggable, and equipped with a "Pause" function. This ensures that the tool acts as a helpful assistant without obstructing the primary meeting interface or consuming unnecessary API quotas.

V. Prototype Showcase

To validate the technical feasibility and user experience of the ASEAN Nuance Decoder, a fully functional prototype was deployed and tested within a live Google Meet environment. The showcase below demonstrates the system's core interface and real-time analytical capabilities.

– **Non-Intrusive Dashboard Interface**:

The prototype features a floating, dark-mode dashboard injected directly into the Google Meet DOM. Designed to minimize cognitive overload, the UI is fully draggable and resizable. It includes a live status indicator and a manual "Pause/Resume" toggle to give users complete control over API consumption during the meeting.

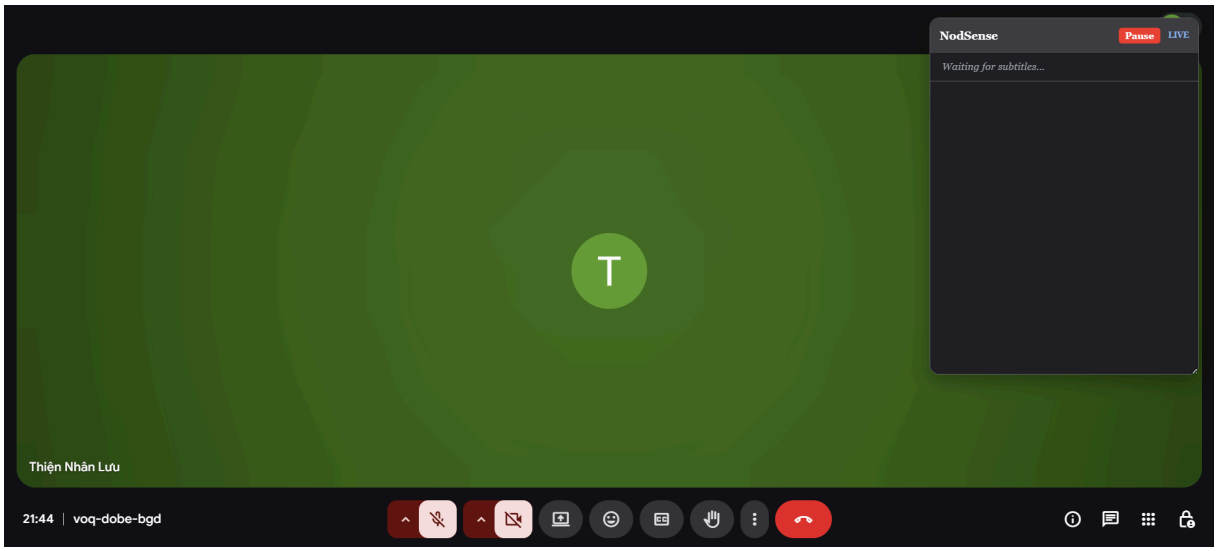

Figure 3: The default idle state of the Nuance Decoder UI within Google Meet.

– **Real-Time Caption Tracking and Keyword Highlighting**:

As the meeting progresses, the **MutationObserver** instantly captures spoken sentences. The prototype successfully isolates the speaker's name and displays the live transcript. To assist the user in quickly grasping the core topic, the system dynamically highlights critical English keywords extracted by the Gemini LLM.

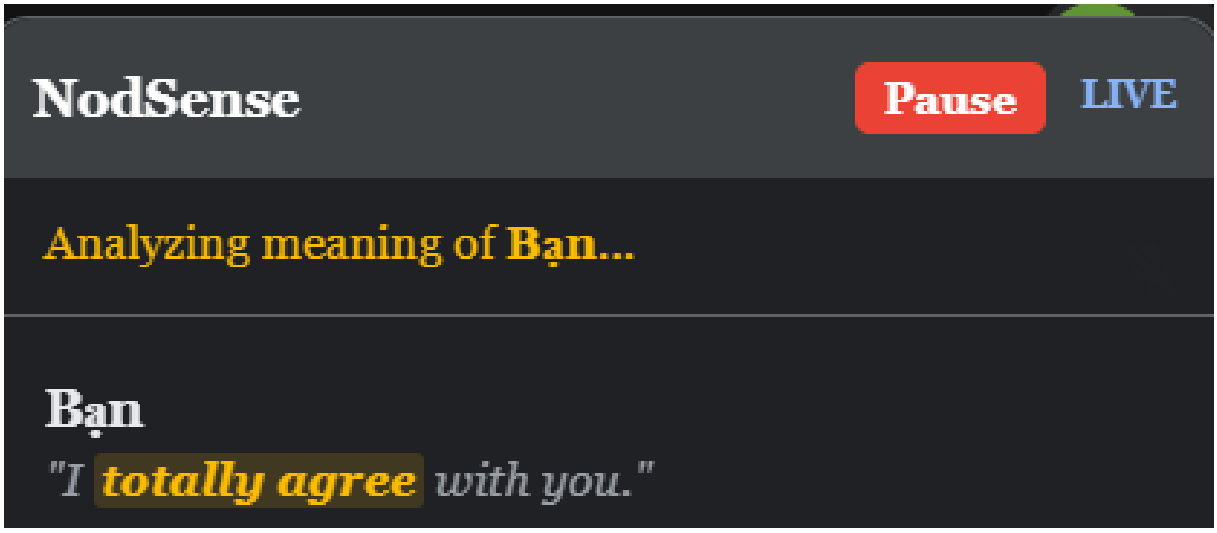

Figure 4: Live transcript tracking and dynamic keyword highlighting during an active conversation.

– **Deep Nuance Analysis and Smart Suggestions**:

The true value of the prototype is demonstrated in its analytical output. Following a 4-second speech pause, the dashboard updates to display the underlying corporate nuance, a direct semantic translation, and quantitative sentiment metrics (Agreement vs. Hesitation percentages). Crucially, the prototype delivers an instant,

context-appropriate English reply suggestion, empowering the user to respond confidently.

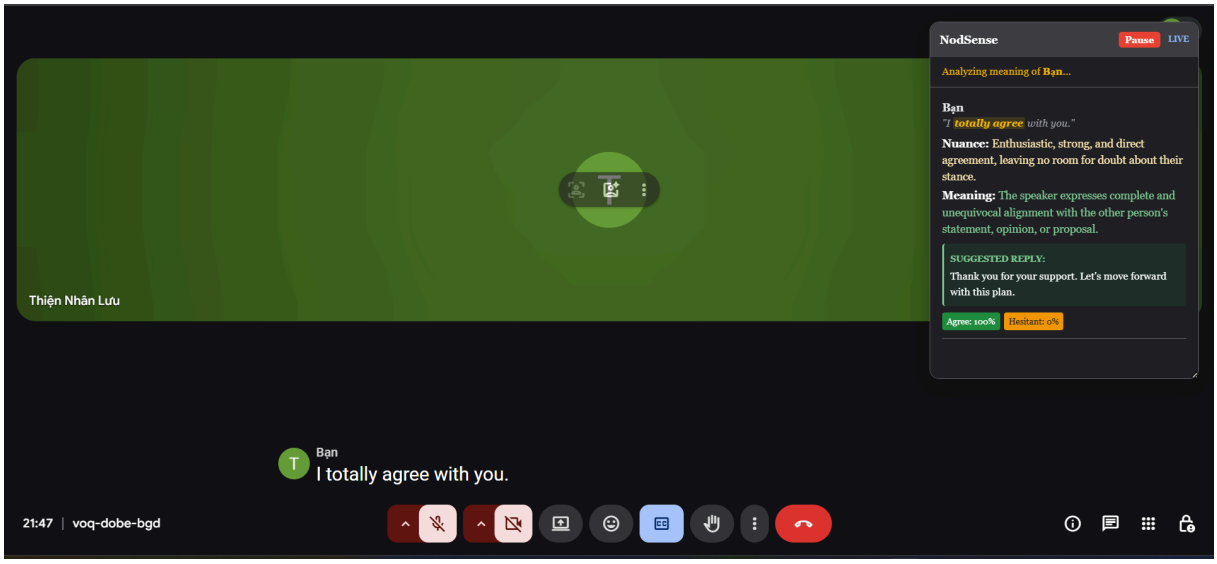

Figure 5: The complete analytical output, providing the user with cultural context and a suggested response.

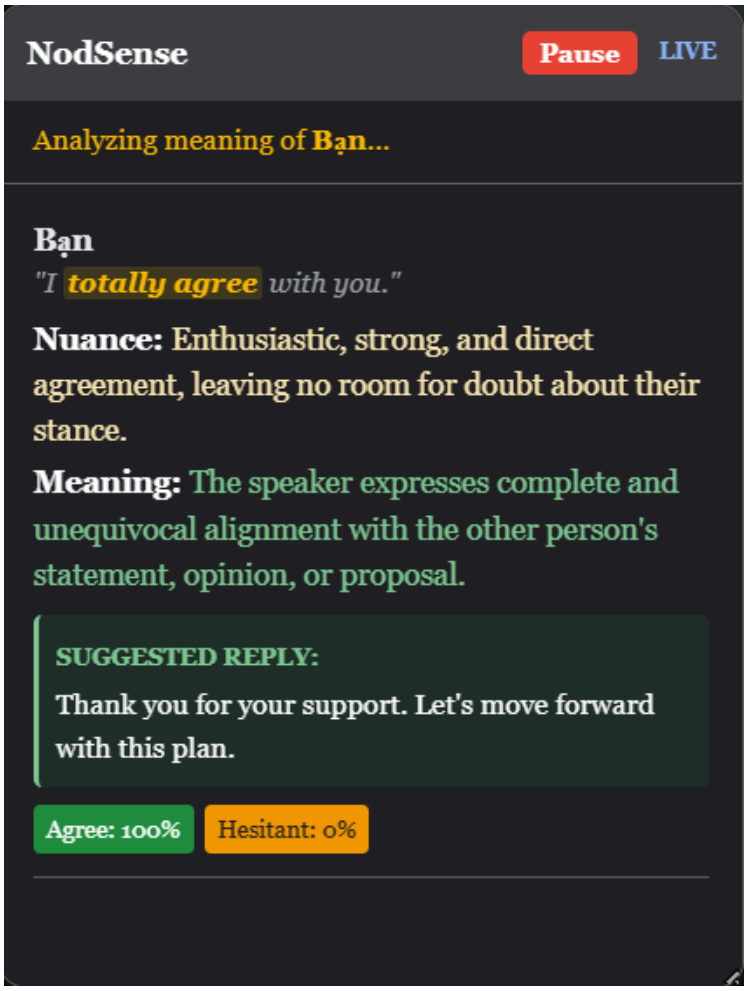

Figure 6: The complete output in dashboard.

VI. AI Acknowledgement

In the development of NodSense, Artificial Intelligence served as both a collaborative partner and a core technical foundation. We leveraged advanced LLM platforms, specifically Google Gemini and ChatGPT, to transition from initial brainstorming to a structured execution plan, while also utilizing these tools to streamline internal team communication and task delegation.

Regarding the technical architecture of our product, we made a strategic decision to prioritize API-based integration over the development of a custom fine-tuned model. While fine-tuning offers deep specialization, it is often a resource-intensive process requiring massive datasets and high-end computational power (GPU hours) that can hinder rapid prototyping.

By utilizing the Gemini 1.5 Flash API, we achieved several key advantages:

High-Performance Context Processing: Access to state-of-the-art reasoning capabilities without the overhead of model maintenance.

Cost-Efficiency: Maintaining a lean budget by paying only for inference, which is crucial for a hackathon-scale project.

Rapid Deployment: Accelerating our time-to-market, allowing the product to be published quickly to target early adopters and gather real-world user feedback for future iterations.

VII. Challenges and Solutions

1. Scalability of Cultural Nuance Mapping

Challenge: Expanding the system beyond a few language pairs (e.g., Vietnamese-Japanese) to hundreds of global cultures is impossible with manual prompting or "hard-coded" rules.

Solution: Developing a Modular Cultural Engine. This engine will autonomously ingest sociological research data and leverage a User Feedback Loop to refine cultural accuracy dynamically through machine learning.

2. Multi-modal Integration (Beyond Text)

Challenge: In real-world communication, 70% of meaning is derived from tone of voice and facial expressions. A text-only extension misses these critical non-verbal cues.

Solution: Expanding into Video Conferencing Plugins (Zoom/Google Meet). Future versions will integrate Prosody Analysis (tone) and Micro-expression Recognition to provide a holistic 360-degree context report.

3. Personalization vs. Cultural Generalization

Challenge: AI can easily confuse an individual's unique personality (being naturally hesitant) with a cultural communication style, leading to inaccurate "stereotyped" analysis.

Solution: Implementing User-Behavioral Baselining. The AI will learn the communication styles of specific frequent contacts to establish a behavioral "baseline," triggering alerts only when there is a significant deviation from their usual pattern.

4. Ethical AI & Bias Mitigation

Challenge: Mislabeling a cultural behavior as "insincere" could reinforce harmful stereotypes or escalate professional conflicts.

Solution: Adopting a "Co-pilot, not Judge" Framework. The AI will present multiple possible scenarios rather than a single definitive conclusion, ensuring that the final social judgment remains in human hands.

VIII. Future Roadmap

Phase 1: Modular Cultural Engine & Platform Expansion: Implementing an autonomous engine that leverages a User Feedback Loop to dynamically refine cultural accuracy through machine learning. Concurrently, the extension will expand beyond Google Meet to support other major conferencing platforms like Zoom.

Phase 2: Multi-Modal Context Integration (Voice & Vision): Evolving the AI from text-only processing to provide a holistic 360-degree context report. This phase introduces Prosody Analysis (tone of voice) and Micro-expression Recognition to capture the critical non-verbal cues that make up 70% of communication.

Phase 3: User-Behavioral Baselining: Shifting from broad cultural generalizations to highly personalized analysis. The AI will learn the unique communication styles of specific frequent contacts, establishing a behavioral baseline to accurately distinguish an individual's natural personality (e.g., being naturally hesitant) from cultural stereotypes.

IX. Conclusion

NodSense effectively addresses cross-cultural miscommunication in global workspaces by acting as a real-time "Cultural Companion" within Google Meet. By leveraging the Gemini API and a Manifest V3 event-driven architecture (utilizing MutationObserver), the extension seamlessly decodes hidden cultural intents, corporate jargon, and tone without disrupting browser performance or exceeding API limits.

Looking ahead, our strategic roadmap prioritizes sustainable, user-led growth, continuous refinement through feedback loops, and an ethical "co-pilot" approach to AI. By providing instant, context-aware insights and reply suggestions, NodSense significantly reduces cognitive load and fosters highly efficient, empathetic global collaboration.

X. Refereneces

[1] N. T. P. Loan, *HIGH – LOW CONTEXT COMMUNICATION: VIETNAMESE STYLE*. Hai Phong language center, 2019.